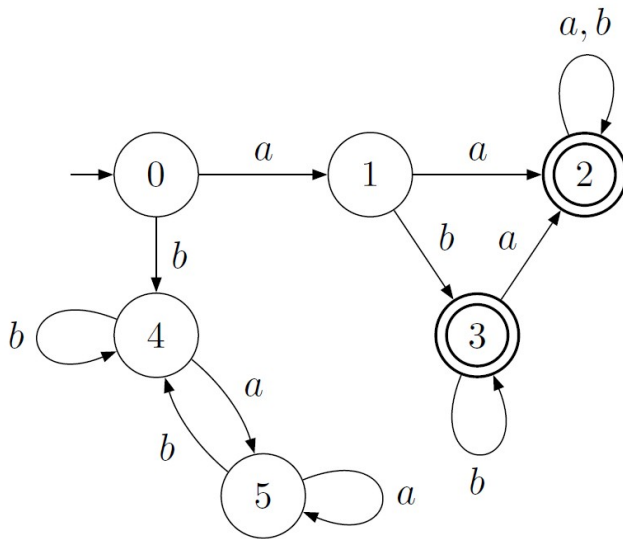




G52LAC (COMP2040)

Languages and Computation

Lecture 7: Pushdown Automata

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Outline

- Recap.
- Pushdown automata (PDA) introduction.
- The language of a PDA.

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- Recap.
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Pushdown Automata vs Finite Automata

- Main limitation of finite automata: **finite memory**
- A Pushdown Automaton is an NFA with an **unlimited** stack and ε -transitions.

$$PDA \approx NFA + \text{Stack} + \varepsilon\text{-transitions}$$

- The stack:
 - infinite memory
 - last in, first out (LIFO)
- ε -transitions:
 - Transitions that can be made without reading an input symbol.
 - They **do not** require the input word to be ε !

PDA Notes

- PDAs are non-deterministic: they accept if there is **any** route to acceptance (like NFAs).
- Every transition **always** pops the top stack symbol.
Thus when the stack is empty, the machine always stops.
If any input remains, the word is rejected (like NFAs).
- The stack may contain symbols that are not part of the input alphabet.
- ϵ -transitions are special transitions that consume no input.
They can be done at any time (provided the stack symbol is correct); the remaining input **does not** have to be empty.

Methods of Acceptance

Pushdown Automata can accept words in one of two ways:

- By Final (or Accepting) State: the automaton is in an accepting state after consuming the input word.
- By Empty (or Null) Stack: the stack is empty after consuming the input word.

Formal Definition of a PDA

- A Final-State PDA is a 7-tuple: $(Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$
- An Empty-Stack PDA is a 6-tuple: $(Q, \Sigma, \Gamma, \delta, q_0, Z_0)$

where

- Q is a finite set of states
- Σ is a finite set of **input** symbols (the input alphabet)
- Γ is a finite set of **stack** symbols (the stack alphabet)
- $\delta \in Q \times (\Sigma \cup \{\varepsilon\}) \times \Gamma \rightarrow \mathcal{P}_{\text{FIN}}(Q \times \Gamma^*)$ is a transition function
- $q_0 \in Q$ is the initial state
- $Z_0 \in \Gamma$ is the initial stack symbol
- $F \subseteq Q$ is the set of accepting states

Here $\mathcal{P}_{\text{fin}}(A)$ are the finite subsets of a set, i.e. this can be defined as

$$\mathcal{P}_{\text{fin}}(A) = \{X \mid X \subseteq A \wedge X \text{ is finite.}\}$$

Recommended Reading

- Introduction to Automata Theory, Languages, and Computation (3rd edition), pages 225–229
- G52MAL Lecture Notes, pages 37–39

Outline

- Recap.
- Pushdown automata (PDA) introduction.
- The language of a PDA.

Recap: Pushdown Automata (PDAs)

- A Final-State PDA is a 7-tuple: $(Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$
- An Empty-Stack PDA is a 6-tuple: $(Q, \Sigma, \Gamma, \delta, q_0, Z_0)$

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- Q is a finite set of states
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$$\mathcal{P}_{\text{fin}}(A) = \{X \mid X \subseteq A \wedge X \text{ is finite.}\}$$

Instantaneous Descriptions (IDs)

- An Instantaneous Description completely describes the state of computation of a PDA.
- Definition: An Instantaneous Description is a triple

$$(q, w, \gamma) \in Q \times \Sigma^* \times \Gamma^*$$

- That is, a state, a word, and a stack configuration.

One-Step Relation

- The relation $i_1 \mid_P i_2$ means:
“PDA P can move in one step from ID i_1 to ID i_2 ”
- Definition:

$$\vdash \subseteq ID \times ID$$

$$1 \quad (q', \alpha) \in \delta(q, x, z) \Rightarrow (q, xw, z\gamma) \vdash (q', w, \alpha\gamma)$$

$$2 \quad (q', \alpha) \in \delta(q, \varepsilon, z) \Rightarrow (q, w, z\gamma) \vdash (q', w, \alpha\gamma)$$

where

$$q, q' \in Q$$

$$x \in \Sigma$$

$$w \in \Sigma^*$$

$$z \in \Gamma$$

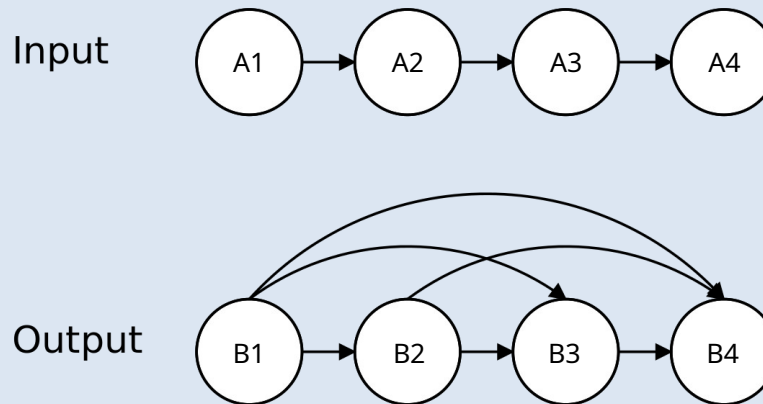
$$\alpha, \gamma \in \Gamma^*$$

Many Steps Relation

- The relation $i_1 \xrightarrow{*}_P i_2$ means:
“PDA P can move in zero or more steps from ID i_1 to ID i_2 ”
- Definition:
 $\xrightarrow{*} \subseteq ID \times ID$ is the reflexive transitive closure of $\xrightarrow{\quad}$

... which roughly means that we can transition from i_1 to i_2 , in zero ore more steps of the one-step relation.

Diagram adapted from a diagram by [Yavoh](#)



Bottom diagram: transitive closure of the relation depicted in the top diagram, e.g. node B1 is connected to B3, through two steps of connections in the top diagram (A1-A2-A3).

Example PDA specification

$$P_1 = (\{0, 1, 2, 3\}, \{a, b\}, \{\#, a\}, \delta, 0, \#, \{0, 3\})$$

where

$$\delta(0, a, \#) = \{(1, a\#)\}$$

$$\delta(1, a, a) = \{(1, aa)\}$$

$$\delta(1, b, a) = \{(2, \varepsilon)\}$$

$$\delta(2, b, a) = \{(2, \varepsilon)\}$$

$$\delta(2, \varepsilon, \#) = \{(3, \varepsilon)\}$$

$$\delta(-, -, -) = \emptyset$$

Example

An example, with details.

As an example we consider a PDA P that recognizes the language of even length palindromes over $\Sigma = \{0, 1\}$ — $L = \{ww^R \mid w \in \{0, 1\}^*\}$. Intuitively, this PDA pushes the input symbols on the stack until it *guesses* that it is in the middle and then it compares the input with what is on the stack, popping of symbols from the stack as it goes. If it reaches the end of the input precisely at the time when the stack is empty, it accepts.

$$P_0 = (\{q_0, q_1, q_2\}, \{0, 1\}, \{0, 1, \#\}, \delta, q_0, \#, \{q_2\})$$

Remember: to show that a PDA accepts a word you only need one accepting sequence. But to show that a PDA rejects a word you have to show that **all** possible sequences reject.

Example

$$P_0 = (\{q_0, q_1, q_2\}, \{0, 1\}, \{0, 1, \#\}, \delta, q_0, \#, \{q_2\})$$

where δ is given by the following equations:

$$\begin{aligned}\delta(q_0, 0, \#) &= \{(q_0, 0\#)\} \\ \delta(q_0, 1, \#) &= \{(q_0, 1\#)\} \\ \delta(q_0, 0, 0) &= \{(q_0, 00)\} \\ \delta(q_0, 1, 0) &= \{(q_0, 10)\} \\ \delta(q_0, 0, 1) &= \{(q_0, 01)\} \\ \delta(q_0, 1, 1) &= \{(q_0, 11)\} \\ \delta(q_0, \epsilon, \#) &= \{(q_1, \#)\} \\ \delta(q_0, \epsilon, 0) &= \{(q_1, 0)\} \\ \delta(q_0, \epsilon, 1) &= \{(q_1, 1)\} \\ \delta(q_1, 0, 0) &= \{(q_1, \epsilon)\} \\ \delta(q_1, 1, 1) &= \{(q_1, \epsilon)\} \\ \delta(q_1, \epsilon, \#) &= \{(q_2, \epsilon)\} \\ \delta(q, w, z) &= \emptyset \quad \text{everywhere else}\end{aligned}$$

Whiteboard

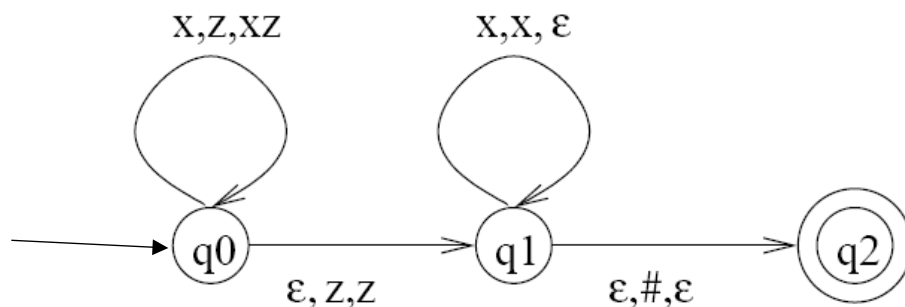
Example

To save space we may abbreviate this by writing:

$$\begin{aligned}\delta(q_0, x, z) &= \{(q_0, xz)\} \\ \delta(q_0, \epsilon, z) &= \{(q_1, z)\} \\ \delta(q_1, x, x) &= \{(q_1, \epsilon)\} \\ \delta(q_1, \epsilon, \#) &= \{(q_2, \epsilon)\} \\ \delta(q, x, z) &= \emptyset \quad \text{everywhere else}\end{aligned}$$

where $q \in Q, x \in \Sigma, z \in \Gamma$. We obtain the previous table by expanding all the possibilities for q, x, z .

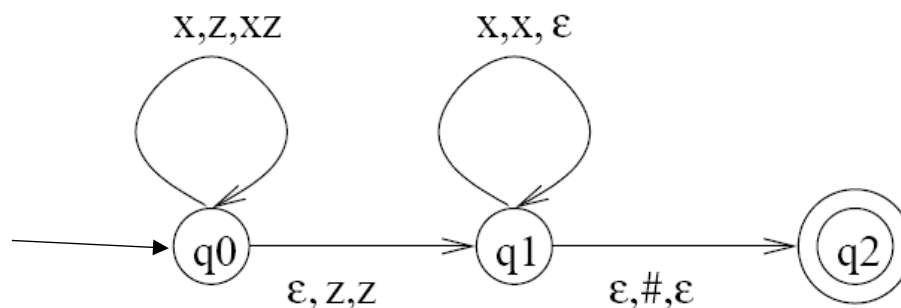
We draw the transition diagram of P by labelling each transition with a triple x, Z, γ with $x \in \Sigma, Z \in \Gamma, \gamma \in \Gamma^*$:



Example

$$\begin{aligned}
 \delta(q_0, 0, \#) &= \{(q_0, 0\#)\} \\
 \delta(q_0, 1, \#) &= \{(q_0, 1\#)\} \\
 \delta(q_0, 0, 0) &= \{(q_0, 00)\} \\
 \delta(q_0, 1, 0) &= \{(q_0, 10)\} \\
 \delta(q_0, 0, 1) &= \{(q_0, 01)\} \\
 \delta(q_0, 1, 1) &= \{(q_0, 11)\}
 \end{aligned}$$

$$\begin{aligned}
 \delta(q_1, 0, 0) &= \{(q_1, \epsilon)\} \\
 \delta(q_1, 1, 1) &= \{(q_1, \epsilon)\}
 \end{aligned}$$



$$\begin{aligned}
 \delta(q_0, \epsilon, \#) &= \{(q_1, \#)\} \\
 \delta(q_0, \epsilon, 0) &= \{(q_1, 0)\} \\
 \delta(q_0, \epsilon, 1) &= \{(q_1, 1)\}
 \end{aligned}$$

$$\delta(q_1, \epsilon, \#) = \{(q_2, \epsilon)\}$$

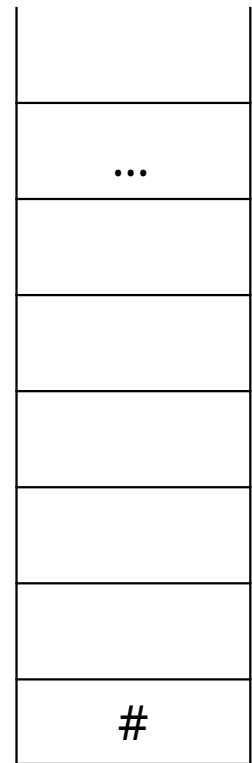
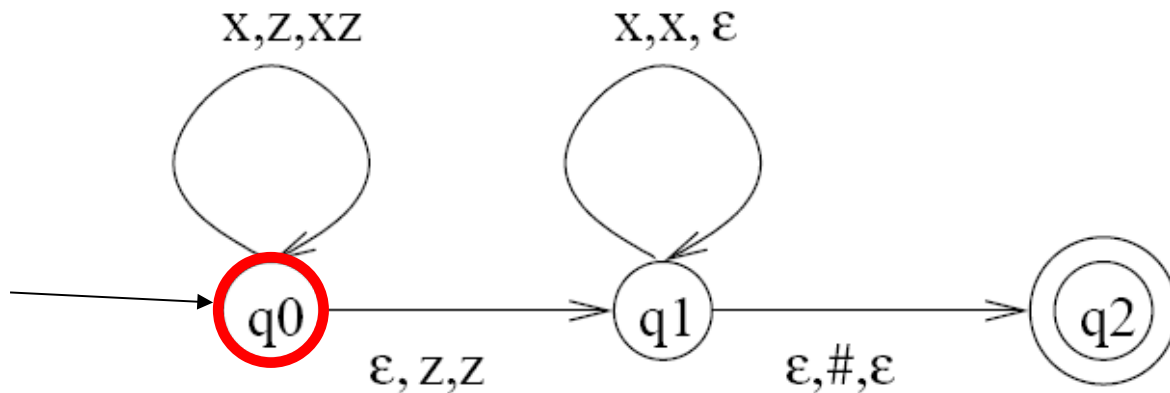
Example

Consider the word 0110 — what are possible sequences of IDs for P_0 starting with $(q_0, 0110, \#)$?

$(q_0, 0110, \#)$

Input word

0110



Stack

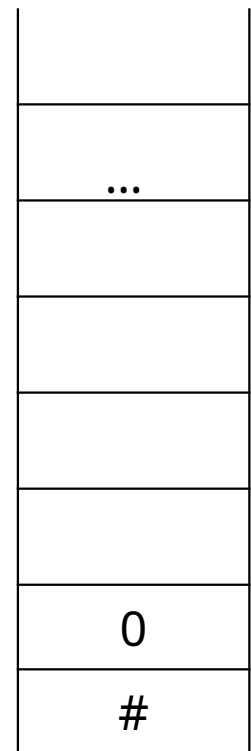
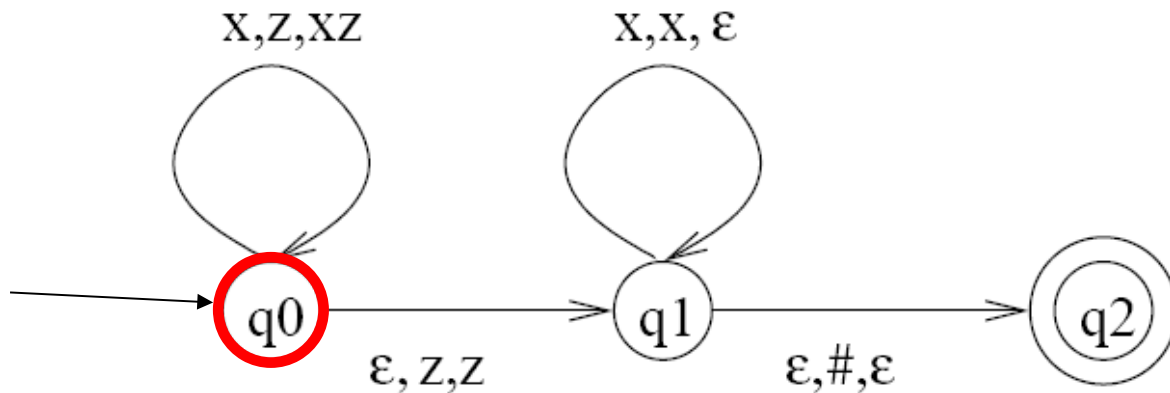
Example

Consider the word 0110 — what are possible sequences of IDs for P_0 starting with $(q_0, 0110, \#)$?

$$(q_0, 0110, \#) \vdash_{P_0} (q_0, 110, 0\#) \quad 1. \text{ with } (q_0, 0\#) \in \delta(q_0, 0, \#)$$

Input word

110



Stack

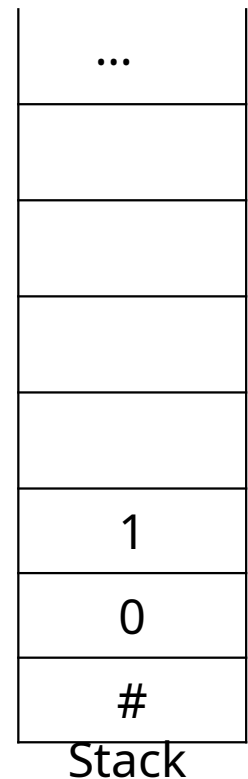
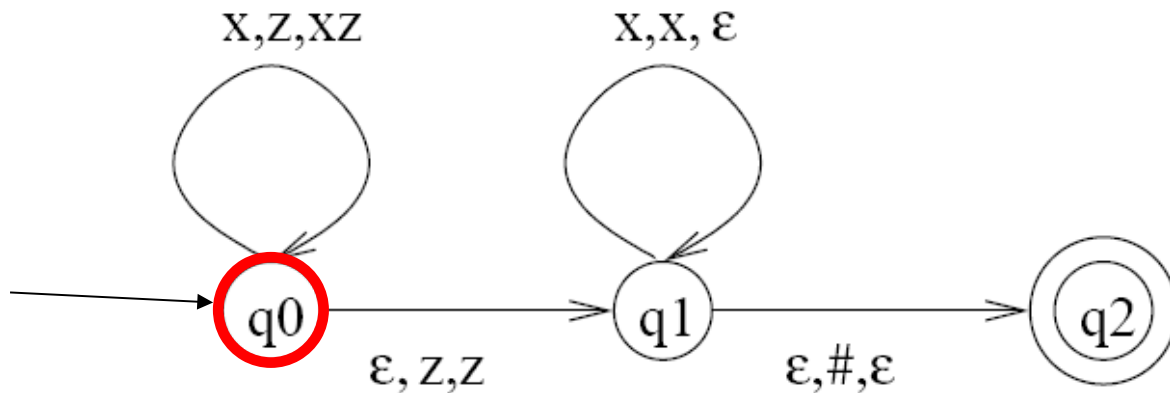
Example

Consider the word 0110 — what are possible sequences of IDs for P_0 starting with $(q_0, 0110, \#)$?

$$\begin{aligned} (q_0, 0110, \#) &\vdash_{P_0} (q_0, 110, 0\#) && 1. \text{ with } (q_0, 0\#) \in \delta(q_0, 0, \#) \\ &\vdash_{P_0} (q_0, 10, 10\#) && 1. \text{ with } (q_0, 10) \in \delta(q_0, 1, 0) \end{aligned}$$

Input word

10



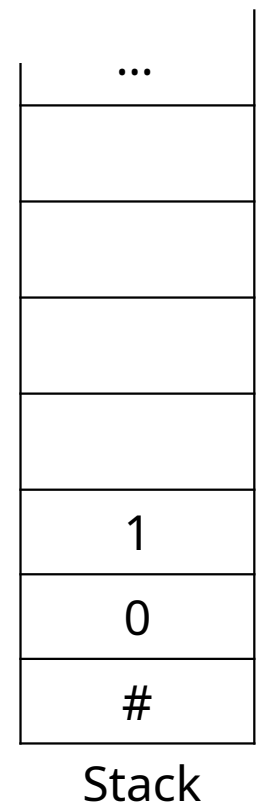
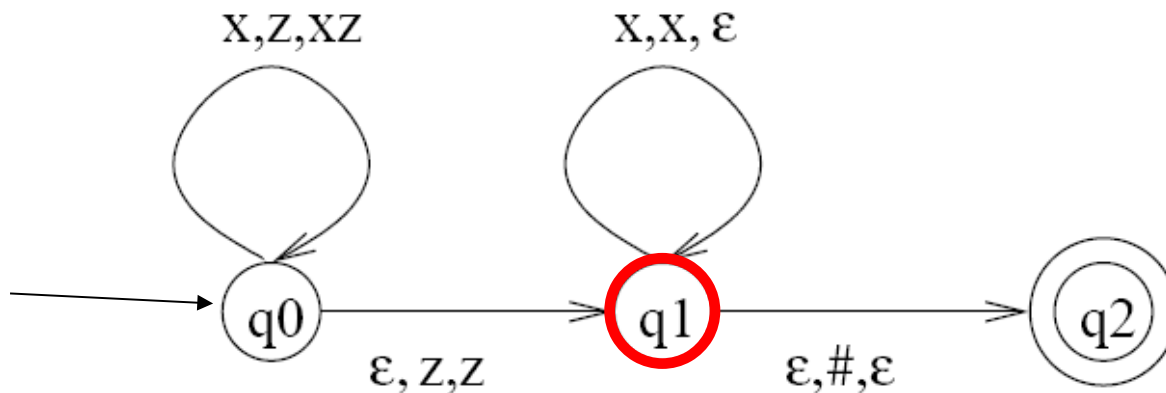
Example

Note. "1" and "2" refer to types of configuration changes (see the first slide on the "one-step relation").

Consider the word 0110 — what are possible sequences of IDs for P_0 starting with $(q_0, 0110, \#)$?

$$\begin{array}{lll}
 (q_0, 0110, \#) & \vdash_{P_0} (q_0, 110, 0\#) & 1. \text{ with } (q_0, 0\#) \in \delta(q_0, 0, \#) \\
 & \vdash_{P_0} (q_0, 10, 10\#) & 1. \text{ with } (q_0, 10) \in \delta(q_0, 1, 0) \\
 & \vdash_{P_0} (q_1, 10, 10\#) & 2. \text{ with } (q_1, 1) \in \delta(q_0, \epsilon, 1)
 \end{array}$$

10



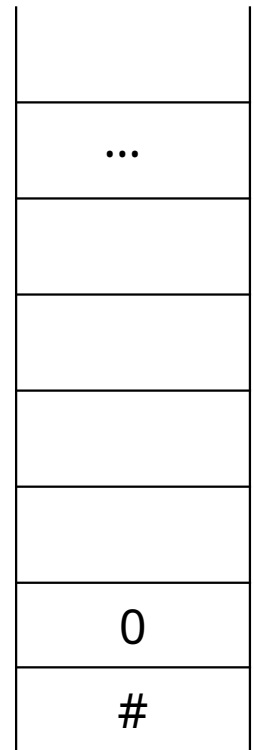
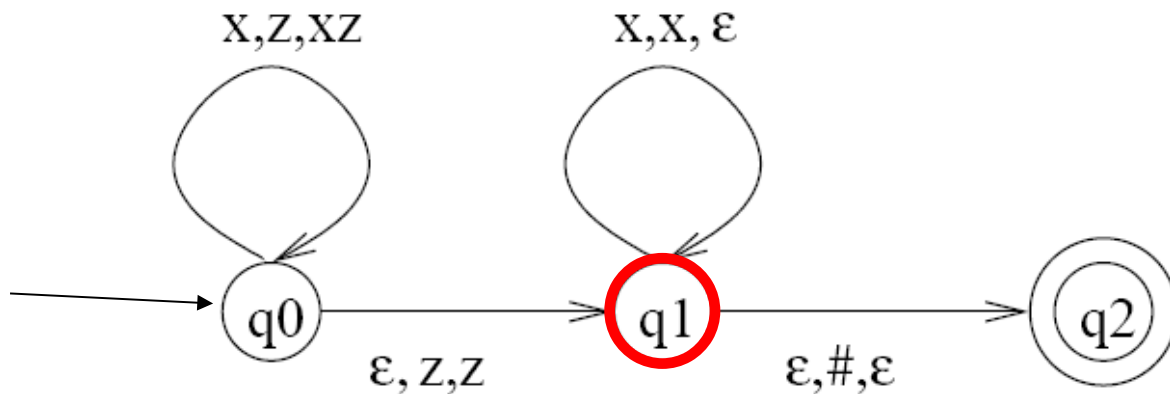
Example

Consider the word 0110 — what are possible sequences of IDs for P_0 starting with $(q_0, 0110, \#)$?

Input word

0

$(q_0, 0110, \#)$	$\vdash_{P_0} (q_0, 110, 0\#)$	1. with $(q_0, 0\#) \in \delta(q_0, 0, \#)$
	$\vdash_{P_0} (q_0, 10, 10\#)$	1. with $(q_0, 10) \in \delta(q_0, 1, 0)$
	$\vdash_{P_0} (q_1, 10, 10\#)$	2. with $(q_1, 1) \in \delta(q_0, \epsilon, 1)$
	$\vdash_{P_0} (q_1, 0, 0\#)$	1. with $(q_1, \epsilon) \in \delta(q_1, 1, 1)$



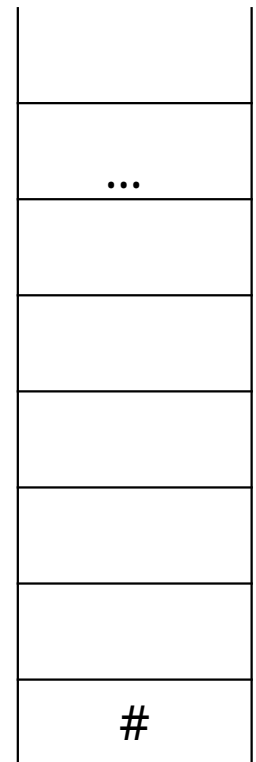
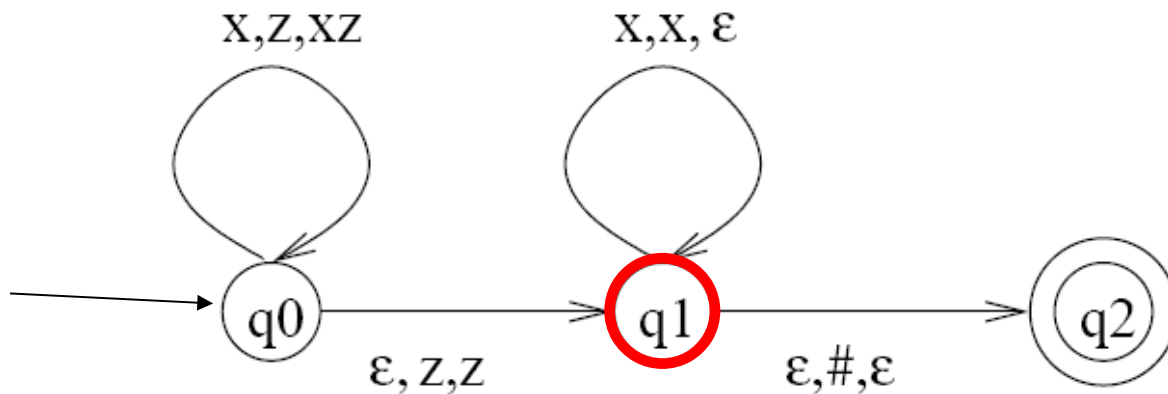
Stack

Example

Consider the word 0110 — what are possible sequences of IDs for P_0 starting with $(q_0, 0110, \#)$?

Input word

$(q_0, 0110, \#)$	$\vdash_{P_0} (q_0, 110, 0\#)$	1. with $(q_0, 0\#) \in \delta(q_0, 0, \#)$
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	$\vdash_{P_0} (q_1, 10, 10\#)$	2. with $(q_1, 1) \in \delta(q_0, \epsilon, 1)$
	$\vdash_{P_0} (q_1, 0, 0\#)$	1. with $(q_1, \epsilon) \in \delta(q_1, 1, 1)$
	$\vdash_{P_0} (q_1, \epsilon, \#)$	1. with $(q_1, \epsilon) \in \delta(q_1, 0, 0)$



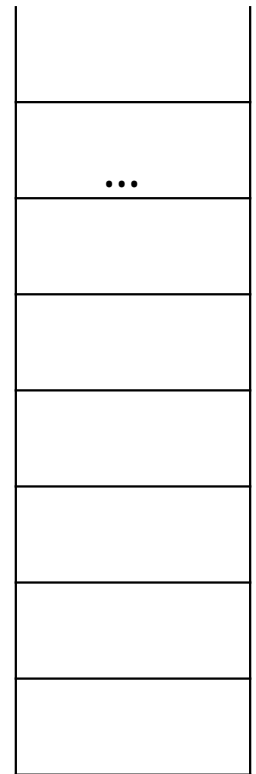
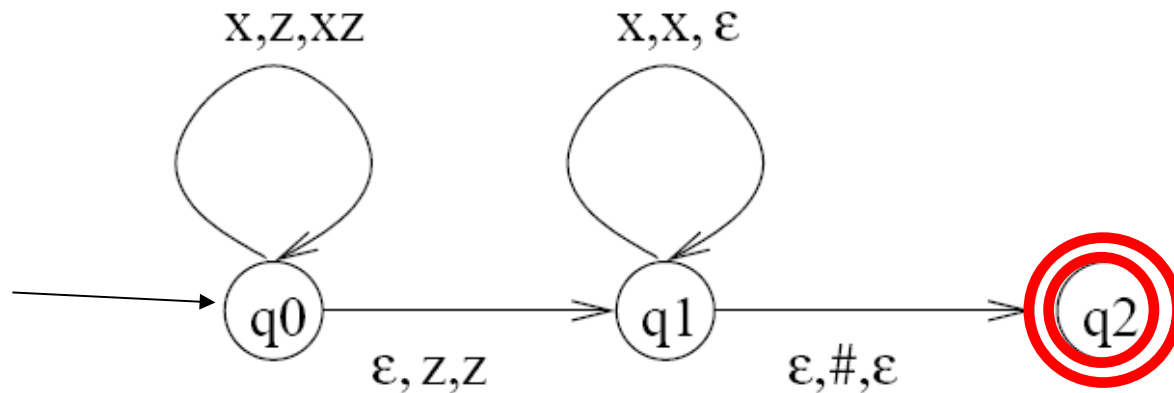
Stack

Example

Consider the word 0110 — what are possible sequences of IDs for P_0 starting with $(q_0, 0110, \#)$?

Input word

$(q_0, 0110, \#)$	$\vdash_{P_0} (q_0, 110, 0\#)$	1. with $(q_0, 0\#) \in \delta(q_0, 0, \#)$
	$\vdash_{P_0} (q_0, 10, 10\#)$	1. with $(q_0, 10) \in \delta(q_0, 1, 0)$
	$\vdash_{P_0} (q_1, 10, 10\#)$	2. with $(q_1, 1) \in \delta(q_0, \epsilon, 1)$
	$\vdash_{P_0} (q_1, 0, 0\#)$	1. with $(q_1, \epsilon) \in \delta(q_1, 1, 1)$
	$\vdash_{P_0} (q_1, \epsilon, \#)$	1. with $(q_1, \epsilon) \in \delta(q_1, 0, 0)$
	$\vdash_{P_0} (q_2, \epsilon, \epsilon)$	2. with $(q_2, \epsilon) \in \delta(q_1, \epsilon, \#)$



Stack

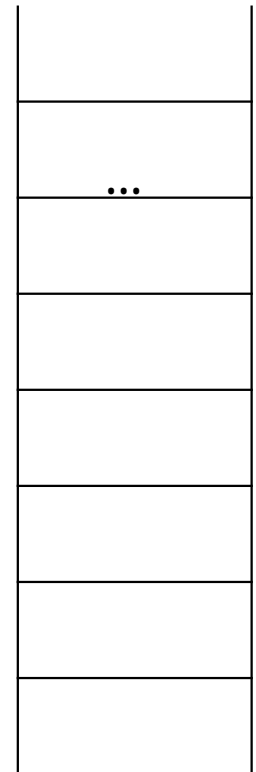
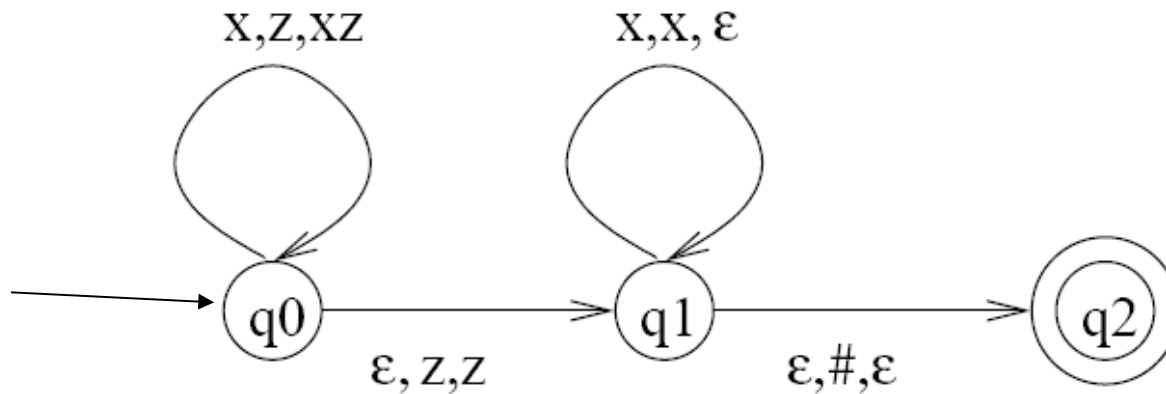
Exercise

How about:

Input word
11

and:

Input word
10



Stack

The Language of a PDA

- Given a Final-State PDA $P = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$, the language of P is defined as:

$$L(P) = \{w \mid ((q_0, w, Z_0) \xrightarrow{*}_P (q, \varepsilon, \gamma)) \wedge (q \in F)\}$$

- Given an Empty-Stack PDA $P = (Q, \Sigma, \Gamma, \delta, q_0, Z_0)$, the language of P is defined as:

$$L(P) = \{w \mid (q_0, w, Z_0) \xrightarrow{*}_P (q, \varepsilon, \varepsilon)\}$$

Recommended Reading

- Introduction to Automata Theory, Languages, and Computation (3rd edition), pages 230–236
- G52MAL Lecture Notes, pages 39–41

Acknowledgements

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